AS/400 → Snowflake Modernization: CIO Executive Briefing Deck

Slide 1 — Title

AS/400 → Snowflake Modernization Program
Enterprise Data Transformation | CIO Executive Briefing
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Slide 2 — Executive Summary

- Legacy **AS/400 (IBM i)** systems limit scalability, analytics, and integration.
- Snowflake provides a **cloud-native, elastic, secure** data platform.
- Migration enables **real-time analytics**, **cost reduction**, and **business agility**.
- Program structured into **8 phases**, delivering a controlled, auditable modernization.
- Zero-defect migration approach with **reconciliation** and **parallel run** ensures business continuity.

Slide 3 — Why Modernize Now

- AS/400 workloads are **costly**, **rigid**, and **difficult to integrate**.
- RPG/CL skills are declining; modernization reduces operational risk.
- Snowflake delivers:
  - Elastic compute & storage
  - Zero-copy cloning
  - Secure data sharing
  - Native support for **DBT**, **AI/ML**, **Python**, **Spark**
- Enables enterprise-wide analytics and digital transformation.

Slide 4 — Target State Architecture

Key Components:

- Cloud Landing Zone (S3/ADLS/GCS)
- Snowflake Stages & Warehouses
- **Snowpipe** for continuous ingestion
- **Streams & Tasks** for CDC
- DBT for transformation & lineage
- BI/AI/ML tools consuming Snowflake

Outcome: A scalable, governed, cloud-native data platform.

Slide 5 — Migration Strategy Overview

- **Wave-based migration** to reduce risk
- **Full load + CDC replay** for accuracy
- **DBT-driven ELT** for transparency
- **Dual-run validation** before cutover
- **Hypercare** to stabilize post-go-live

Slide 6 — Migration Phases

1. **Initiation & Architecture**
2. **AS/400 Discovery & Reverse Engineering**
3. **Extraction Framework (Full + CDC)**
4. **Landing Zone & Staging**
5. **Snowflake Ingestion**
6. **Transformation (DBT)**
7. **Validation & Reconciliation**
8. **Cutover & Hypercare**

Each phase includes governance, QA, and business checkpoints.

Slide 7 — AS/400 Discovery Highlights

- **PF/LF inventory**

- **RPG III/IV logic analysis**
- Job scheduler mapping
- Data profiling (packed, zoned, EBCDIC)
- CDC feasibility assessment

Outcome: Complete understanding of legacy logic and data structures.

Slide 8 — Extraction & Ingestion Approach

Extraction:

- DB2/400 ODBC/JDBC
- Fixed-width/EBCDIC exports
- Journaling-based CDC

Ingestion:

- **COPY INTO** for batch loads
- **Snowpipe** for continuous ingestion
- **Streams & Tasks** for incremental merges

Outcome: Reliable, repeatable, auditable ingestion pipelines.

Slide 9 — Transformation Approach (DBT)

- Staging → Intermediate → Marts
- Rebuild RPG logic in SQL/DBT
- SCD Type 1/2 support
- Automated testing & lineage
- CI/CD integration

Outcome: Transparent, governed, modular transformation framework.

Slide 10 — Validation & Reconciliation

- **Schema validation**
- **Metric validation**

- **Row-level reconciliation**
- **Functional validation**

Outcome: Zero-defect migration with full auditability.

Slide 11 — Cutover Strategy

- Freeze AS/400
- Final full load
- Final CDC replay
- Parallel run
- Business sign-off
- Switch downstream systems to Snowflake

Outcome: Controlled, low-risk transition.

Slide 12 — Benefits to the Enterprise

- **Cost Reduction:** Lower infra + operational overhead
- **Scalability:** Elastic compute for peak workloads
- **Speed:** Faster analytics, faster insights
- **Governance:** Centralized RBAC/ABAC
- **Innovation:** AI/ML-ready platform

Slide 13 — Risks & Mitigations

- RPG logic complexity → **Deep discovery + SME validation**
- CDC gaps → **Journaling review + Streams & Tasks**
- Data quality issues → **Multi-layer reconciliation**
- Performance tuning → **Clustering + warehouse sizing**

Outcome: Risks actively managed throughout program.

Slide 14 — Program Timeline (12 Weeks)

- Weeks 1–2: Architecture & Discovery
- Weeks 2–4: Reverse Engineering
- Weeks 4–6: Extraction Framework
- Weeks 6–8: Landing + Ingestion
- Weeks 8–10: Transformation
- Weeks 10–11: Validation
- Weeks 11–12: Cutover & Hypercare

Slide 15 — Investment & ROI

Investment Areas:

- Cloud platform
- Engineering resources
- DBT & automation
- Validation & QA

ROI:

- Reduced legacy maintenance
- Faster analytics
- Lower operational risk
- Improved compliance & auditability

Slide 16 — Final Recommendation

Proceed with the AS/400 → Snowflake modernization to:

- Reduce technical debt
- Enable enterprise analytics
- Improve operational resilience
- Support AI/ML initiatives
- Future-proof the data platform

Snowflake becomes the enterprise system of record for analytics.